

Po-Wei (Brian) Chen

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EDUCATION

Soochow University, Taipei, Taiwan

B.B.A. in Financial Engineering and Actuarial Mathematics

Sept. 2022 – June 2025

- Graduated one year early (met criteria of full credit completion and top 10% class rank each semester)
- GPA 4.0/4.0; Average 95.2/100; Class Rank 1st/53

WORK EXPERIENCE

Research Assistant (Full-time)

Institute of Statistical Science, Academia Sinica, Taipei, Taiwan

Aug. 2025 – present

PI: Prof. Hsin-Cheng Huang, Distinguished Research Fellow

- Augmented diffusion-based generative models with catalytic priors that incorporate information from simpler statistical models to guide training and improve performance when only limited observations are available.
- Assisted in conference operations for the Joint Meetings of the 2025 Taipei International Statistical Symposium and the 13th ICSA International Conference.

High School Math Tutor

Aug. 2024 – June 2025

Taiwan TACT Education (Private Tutoring Center), Minsheng Branch, Taipei, Taiwan

- Taught high school mathematics and prepared students for university entrance examinations.

HONORS & AWARDS

ISI-ISM-ISSAS Joint Conference Poster Competition Award (2026)

Title: Bayesian Inference for Diffusion Models: Integrating Synthetic and Observed Data with Catalytic Priors.

Outstanding Graduating Student (2025)

Soochow University, Taipei, Taiwan

Academic Excellence Award (2022–2025; awarded every semester to the top three out of a class of 50+ students)

Soochow University, Taipei, Taiwan

RESEARCH PROJECTS

Bridging Synthetic and Real Data with Catalytic Priors for Conditional Generative Models

Institute of Statistical Science, Academia Sinica, Taipei, Taiwan

- Developed a novel diffusion-based framework augmented with catalytic priors that leverage simpler statistical structures to guide learning in data-scarce settings.

Empirical Asset Pricing via Machine Learning for Cryptocurrency

Soochow University, Taipei, Taiwan

- Evaluated various machine learning models for factor-based quantile investment strategies. Demonstrated the effectiveness of machine learning in cryptocurrency factor investing.

CONFERENCE PRESENTATIONS

- “**Bridging Synthetic and Real Data with Catalytic Priors for Conditional Generative Models**” 14th IASC-ARS & SoAI Joint Conference, Chiang Mai University, Chiang Mai, Thailand. December, 2026 [Accepted for Oral Presentation]
- “**Bayesian Inference for Diffusion Models: Integrating Synthetic and Observed Data with Catalytic Priors**” ISI-ISM-ISSAS Joint Conference, Academia Sinica, Taipei, Taiwan. February, 2026 [Poster Presentation]
- “**Empirical Asset Pricing via Machine Learning for Cryptocurrency**” Annual Conference and International Symposium of the Taiwan Risk and Insurance Association, Taipei, Taiwan. December, 2025 [Oral Presentation]

COURSEWORK & CERTIFICATIONS

Relevant Coursework

Advanced Calculus, Calculus(I)(II), Linear Algebra(I)(II), Probability and Statistics(I)(II), Advanced Statistics, Regression Analysis, Statistical Inference, Applied Statistics, Numerical Analysis, Operation Research

Open Online Courses

- Python for Everybody Specialization, University of Michigan (via Coursera)
- Programming for Business Computing in Python (1), National Taiwan University (via Coursera)

Exam P (Probability), Society of Actuaries (SOA)

SKILLS & TECHNICAL

Programming: Python, R (Markdown, Shiny), C++ , SQL

Machine Learning Packages: PyTorch, TensorFlow, scikit-learn

EXTRACURRICULAR ACTIVITIES

Statistics Cup Basketball Tournament Champions

2025

Institute of Statistical Science, Academia Sinica, Taipei, Taiwan

Vice Captain of Basketball Team

2024 – 2025

Department of Financial Engineering and Actuarial Mathematics, Soochow University, Taipei, Taiwan

- Led the team to six championships in inter-departmental competitions.